First-Order System Response

One pole, one time constant, no overshoot.

Ogata, *Modern Control Engineering*, Ch. 5.

In this tutorial you will:

- derive the $1/(Ts+1)$ step response and the 63.2% rule,
- read the time constant T off the initial slope, and
- see the ramp and impulse responses of a first-order lag.

Step through with **Ctrl+Enter**, or render a report with `publish`.

A first-order system has the transfer function

$\frac{Y(s)}{R(s)} = \frac{1}{Ts+1}$

where T is the system's **time constant**. Physically this is the same form derived for the RC circuit, the liquid-level tank, and the thermometer in `Mathematical Models/`.

Contents

Unit-Step Response Derivation

With $R(s) = 1/s$:

$Y(s) = \frac{1}{Ts+1} \cdot \frac{1}{s} = \frac{1}{s} - \frac{T}{Ts+1}$

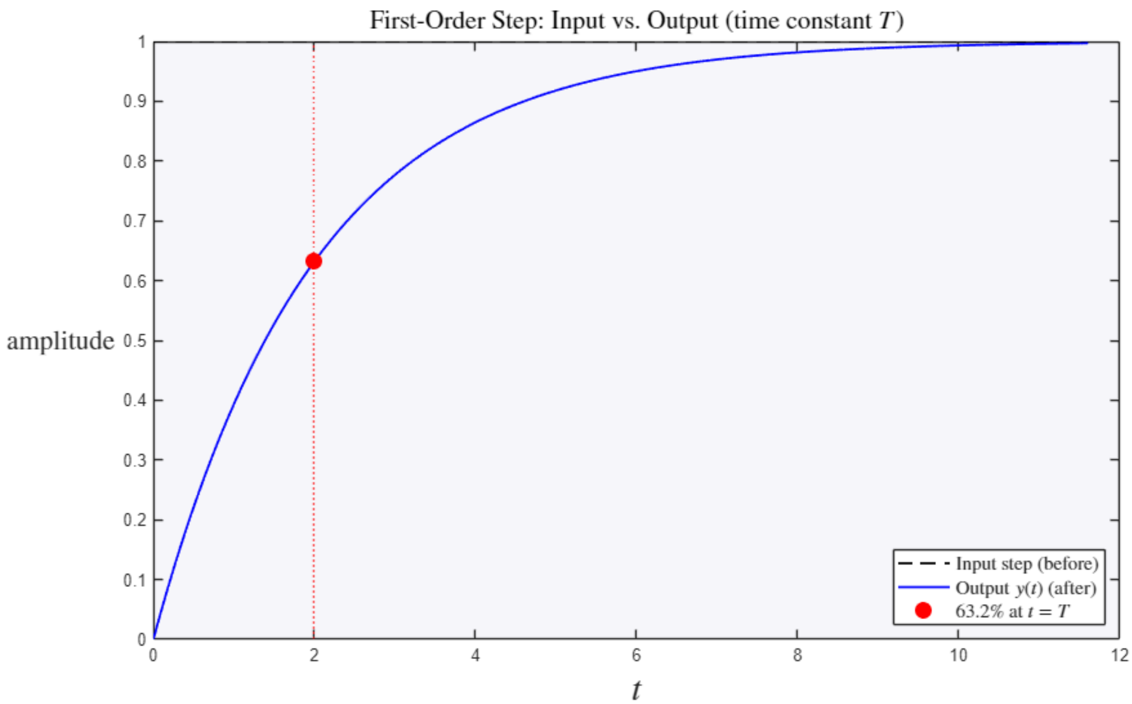
Taking the inverse Laplace transform:

$y(t) = 1 - e^{-t/T}, \quad t \geq 0$

so $y(t) \to 1$ as $t \to \infty$ with no overshoot – a first-order system's step response is always monotonic.

```
T = 2;
G = tf(1,[T 1]);

figure
[ys,ts] = step(G);
plot(ts, ones(size(ts)),'k--', ts, ys,'b','LineWidth',1.3)
hold on
plot(T, 1-exp(-1),'ro','MarkerSize',9,'MarkerFaceColor','r')
xline(T,'r:')
hold off
legend('Input step (before)','Output $y(t)$ (after)','$63.2\%$ at $t=T$', ...
    'Interpreter','latex','FontSize',11,'Location','southeast')
title('First-Order Step: Input vs. Output (time constant $T$)','Interpreter','latex','FontSize',15)
ylabel('amplitude','Interpreter','latex','FontSize',16)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$','Interpreter','latex','FontSize',20)
```



The 63.2% Time-Constant Rule

At $t = T$: $y(T) = 1 - e^{-1} \approx 0.632$. At $t = 2T$, $y(2T) = 1 - e^{-2} \approx 0.865$; at $t = 3T$, $y(3T) = 1 - e^{-3} \approx 0.950$; at $t = 4T$, $y(4T) \approx 0.982$. By convention the system is considered to have settled after about $4T$ (the 2% settling-time rule).

```
[y,t] = step(G);
y_at_T  = interp1(t,y,T);
y_at_2T = interp1(t,y,2*T);
y_at_3T = interp1(t,y,3*T);
y_at_4T = interp1(t,y,4*T);
fprintf('y(T)  = %.4f (expect ~0.632)\n', y_at_T)
fprintf('y(2T) = %.4f (expect ~0.865)\n', y_at_2T)
fprintf('y(3T) = %.4f (expect ~0.950)\n', y_at_3T)
fprintf('y(4T) = %.4f (expect ~0.982)\n', y_at_4T)
```

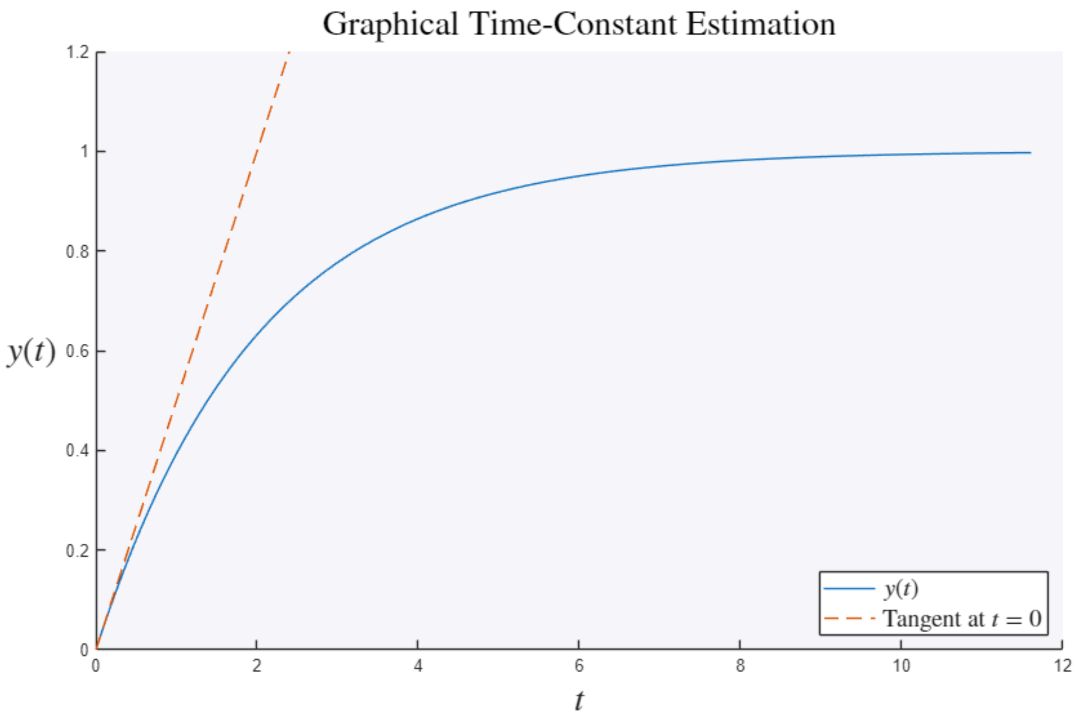
```
y(T)  = 0.6320 (expect ~0.632)
y(2T) = 0.8646 (expect ~0.865)
y(3T) = 0.9502 (expect ~0.950)
y(4T) = 0.9817 (expect ~0.982)
```

Slope at the Origin

Differentiating $y(t) = 1 - e^{-t/T}$ gives $\dot{y}(0) = 1/T$: the tangent line drawn at $t = 0$ reaches the final value $y = 1$ exactly at $t = T$, a graphical method for reading T off an experimental step response.

```
tangent = t/T;
figure
hold on
plot(t,y)
plot(t,tangent,'--')
ylim([0 1.2])
hold off
legend('$y(t)$','Tangent at $t=0$','Interpreter','latex','FontSize',14,'Location','southeast')
title('Graphical Time-Constant Estimation','Interpreter','latex','FontSize',20)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
```

```
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$','Interpreter','latex','FontSize',20)
```

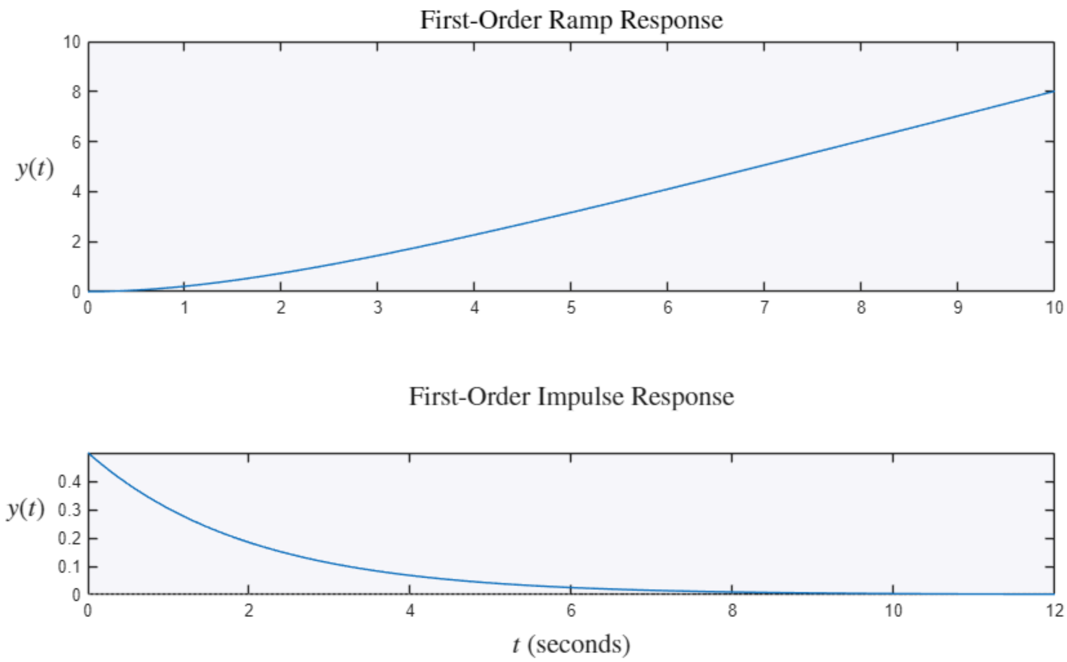


Unit-Ramp and Unit-Impulse Response

For a unit ramp input, $Y(s) = \frac{1}{s^2(Ts+1)}$, giving $y(t) = t - T + Te^{-t/T}$; the steady-state error to a ramp is a constant lag of T seconds. For a unit impulse, $y(t) = \frac{1}{T}e^{-t/T}$ – a decaying exponential starting at $1/T$.

```
figure
subplot(2,1,1)
t2 = 0:0.01:10;
y_ramp = t2 - T + T*exp(-t2/T);
plot(t2,y_ramp)
title('First-Order Ramp Response','Interpreter','latex','FontSize',16)
ylabel('$y(t)$','Interpreter','latex','FontSize',16)
set(get(gca, 'YLabel'), 'Rotation', 0)

subplot(2,1,2)
impulse(G)
title('First-Order Impulse Response','Interpreter','latex','FontSize',16)
ylabel('$y(t)$','Interpreter','latex','FontSize',16)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$','Interpreter','latex','FontSize',16)
```



Try it yourself

- Change the time constant `T` from 2 to 0.5 and notice every marker (63%, tangent, settling) rescale in time while the final value stays 1.
- Read `T` off the plot as the time where the tangent at the origin hits 1.